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29. I write now

$$\alpha + \beta = \sigma, \quad r - 2\delta - \sigma = \beta,$$

and I consider the expression

$$D^r H - \frac{n-1}{1} D^r H_1 + \&c.;$$

the general term of which is

$$R_r^\delta \{ \delta + 1, \delta + 2 + s, (r - 2\delta + 1 - s) \} \\ \times \left\{ \begin{array}{l} \frac{[n-1]^\delta}{1} [n-2]^\sigma [n-2]^\beta \\ - \frac{n-1}{1} [n-2]^\sigma [n-2]^\beta \\ + \frac{n-1 \cdot n-2}{1 \cdot 2} [n-3]^\sigma [n-3]^\beta \\ + \&c. \end{array} \right\},$$

or, as this may be written, putting $q = n - \delta - 1$,

$$R_r^\delta \{ \delta + 1, \delta + 2 + s, (r - 2\delta + 1 - s) \} \\ \times [n-1]^\delta \left\{ \begin{array}{l} [n-2]^\sigma [n-2]^\beta \\ - \frac{q}{1} [n-3]^\sigma [n-3]^\beta \\ + \&c. \end{array} \right\}.$$

30. I assume $r \not\geq n - 2$, we have then $\alpha + \beta = r - \delta \not\geq n - \delta - 2$, and therefore $\alpha + \beta < q$. The general term of the series in $\{ \}$ is

$$(-)^\xi \frac{[q]^\xi}{[\xi]^\xi} [n-2-\xi]^\sigma [n-2-\xi]^\beta,$$

where the terms for which $n - 2 - \xi$ is negative are to be excluded, or what is the same thing, the series is not to be continued beyond $\xi = n - 2$. But observing that $[q]^\xi$ vanishes for $\xi > q$, that is, $\xi > n - \delta - 1$, it is in fact the same thing whether the series is continued indefinitely or only to the term for which $\xi = n - \delta - 1$, and we may consistently with the condition $\xi \not\geq n - 2$, continue the series as far as $\xi = n - \delta - 1$, except in the case $\delta = 0$, when by doing so we include the term corresponding to $\xi = n - 1$, which in virtue of the condition ought to be excluded. The expression for the term in question is $(-)^{n-1}[-1]^\sigma[-1]^\beta$; hence if the sum of the series continued to the proper point is S , the sum continued indefinitely (in the particular case $\delta = 0$) is $S + (-)^{n-1}[-1]^\sigma[-1]^\beta$, but in every other case the sum continued indefinitely is simply S . And by a well-known theorem in finite differences, the sum continued indefinitely is in fact zero. That is, except in the case $\delta = 0$, we have $S = 0$, but in the excepted case

$$S + (-)^{n-1}[-1]^\sigma[-1]^\beta = 0;$$

or observing that $\alpha + \beta (= r - \delta)$ is in this case $= r$, and transforming the factorials, we have

$$S = (-)^{n-r}[\alpha]^\alpha[\beta]^\beta,$$

or substituting for α and β their values,

$$S = (-)^{n-r}[s]^s[r-s]^{r-s}.$$

31. Hence the general term of

$$D^r H - \frac{n-1}{1} D^r H_1 + \&c.$$

vanishes except for $\delta = 0$, but when $\delta = 0$, its value is

$$R_r^s \{1, 2+s, (r+1-s)\} \times (-)^{n-r} [s]^s [r-s]^{r-s};$$

or observing that R_r^s is equal to $[r]^r \div [s]^s [r-s]^{r-s}$, the value is simply

$$(-)^{n-r} [r]^r \{1, 2+s, (r+1-s)\},$$

that is, we have

$$\Delta^2 \left(D^r H - \frac{n-1}{1} D^r H_1 + \&c. \dots \right) = (-)^{n-r} [r]^r \Sigma \{1, 2+s, (r+1-s)\},$$

the summation in respect to s extending from $s = 0$ to $s = r$. In particular, giving to r the values $n-2$ and $n-1$, and attending to the expressions for III and IV, we find

$$\Delta^2 \left(D^{n-2} H - \frac{n-1}{1} D^{n-2} H_1 + \&c. \dots \right) = -[n-2]^{n-2} \text{III},$$

$$\Delta^2 \left(D^{n-1} H - \frac{n-1}{1} D^{n-1} H_1 + \&c. \dots \right) = -[n-3]^{n-1} \text{IV}.$$

32. The equation $\text{III} = 0$ belongs to the curve which by its intersections with the tangent, gives the tangentials of a point of the curve $U = 0$. Hence the equation of the curve in question is

$$D^{n-2} H - \frac{n-1}{1} D^{n-2} H_1 + \&c. = 0,$$

which is Mr Salmon's theorem, leading to the solution of the problem of double tangents.

33. The expressions for I and II are obtained from those of IV and III by interchanging (X, Y, Z) and (x, y, z) , and reversing the sign. Hence if, as before, $\mathfrak{H}$, $\mathfrak{D}$, &c. denote the values which H , D , &c. assume by this interchange, we have

$$\Delta^2 \left(\mathfrak{D}^{n-2} \mathfrak{H} - \frac{n-1}{1} \mathfrak{D}^{n-2} \mathfrak{H}_1 + \&c. \dots \right) = [n-2]^{n-2} \text{II},$$

$$\Delta^2 \left(\mathfrak{D}^{n-1} \mathfrak{H} - \frac{n-1}{1} \mathfrak{D}^{n-1} \mathfrak{H}_1 + \&c. \dots \right) = [n-3]^{n-1} \text{I},$$

and the identical equation

$$a_0 \text{I} + a_1 \text{II} + a_{n-1} \text{III} + a_n \text{IV} = 0$$

becomes therefore

$$\left\{ \begin{array}{l} \left(\mathfrak{D}^{n-1}\mathfrak{H} - \frac{n-1}{1} \mathfrak{D}^{n-1}\mathfrak{H}_1 + \&c. \right) U \\ + \frac{1}{n(n-2)} \left(\mathfrak{D}^{n-2}\mathfrak{H} - \frac{n-1}{1} \mathfrak{D}^{n-2}\mathfrak{H}_1 + \&c. \right) DU \\ - \frac{1}{n(n-2)} \left(D^{n-2}H - \frac{n-1}{1} D^{n-2}H_1 + \&c. \right) \mathfrak{D}\mathbf{T} \\ - \left(D^{n-1}H - \frac{n-1}{1} D^{n-1}H_1 + \&c. \right) \mathbf{T} \end{array} \right\} = 0,$$

which is the general identical equation referred to in the introduction to the present memoir.

34. It is to be noticed that for $n = 3$, the equation is

$$(\mathfrak{D} - 2\mathfrak{H}_1)U + \frac{1}{3}\mathfrak{D}\mathfrak{H} \cdot DU - \frac{1}{3}DH \cdot \mathfrak{D}\mathbf{T} - (H - 2H_1)\mathbf{T} = 0.$$

But we have $H_1 = H$, and in like manner $\mathfrak{H}_1 = \mathfrak{H}$, and the equation thus becomes

$$-\mathfrak{H}U + \frac{1}{3}\mathfrak{D}\mathfrak{H} \cdot DU - \frac{1}{3}DH \cdot \mathfrak{D}\mathbf{T} + H\mathbf{T} = 0.$$

And so also for $n = 4$, the equation is

$$(\mathfrak{D}^2\mathfrak{H} - 3\mathfrak{D}^2\mathfrak{H}_1)U + \frac{1}{8}(\mathfrak{D}^2\mathfrak{H} - 3\mathfrak{D}^2\mathfrak{H}_1)DU - \frac{1}{8}(D^2H - 3D^2H_1)\mathfrak{D}\mathbf{T} - (DH - 3DH_1)\mathbf{T} = 0.$$

But we have in general $DH_1 = \frac{n-3}{n-2}DH$, and therefore in the present case $DH_1 = \frac{1}{2}DH$, and consequently $\mathfrak{D}\mathfrak{H}_1 = \frac{1}{2}\mathfrak{D}\mathfrak{H}$, and the equation thus becomes

$$-\mathfrak{D}\mathfrak{H} \cdot U + \frac{1}{8}(\mathfrak{D}^2\mathfrak{H} - 3\mathfrak{D}^2\mathfrak{H}_1)DU - \frac{1}{8}(D^2H - 3D^2H_1)\mathfrak{D}\mathbf{T} + DH \cdot \mathbf{T} = 0,$$

which agree with the results previously obtained for the two particular cases.

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ON THE CONIC OF FIVE-POINTIC CONTACT AT ANY POINT OF A PLANE CURVE.

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The tangent is a line passing through two consecutive points of a plane curve, and we may in like manner consider the conic which passes through five consecutive points of a plane curve; and as there are certain singular points, viz. the points of inflexion where three consecutive points of the curve lie in a line, so there are singular points where six consecutive points of the curve lie in a conic. In the particular case where the given curve is a cubic, the last-mentioned species of singular points have been considered by Plücker and Steiner, and in the same particular case, the theory of the conic of five-pointic contact has recently been established by Mr Salmon. But the general case, where the curve is of any order whatever, has not so far as I am aware been hitherto considered;—the establishment of this theory is the object of the present memoir.

I. INVESTIGATION OF THE EQUATION OF THE CONIC OF FIVE-POINTIC CONTACT.

1. I take (X, Y, Z) as current coordinates, and I represent the equation of the given curve by

$$\mathbf{T} = (* \mid X, Y, Z)^m = 0.$$

Let (x, y, z) be the coordinates of a given point on the curve, and let $U = (* \mid x, y, z)^m$ be what $\mathbf{T}$ becomes when (x, y, z) are written in the place of (X, Y, Z) ; we have therefore $U = 0$ as a condition satisfied by the coordinates of the point in question.

2. Write for shortness

$$DU = (X\partial_x + Y\partial_y + Z\partial_z)U,$$

$$D^2U = (X\partial_x + Y\partial_y + Z\partial_z)^2U,$$

and let $\Pi = aX + bY + cZ = 0$ be the equation of a line. It is easy to see that

$$D^2U - \Pi \cdot DU = 0$$

will be the equation of a conic having an ordinary (two-pointic) contact with the curve at the point (x, y, z) . In fact the equation $DU = 0$ is that of the tangent at the point in question, and the equation $D^2U = 0$ is that of the penultimate polar (or polar conic) of the point, which conic is touched by the tangent; the assumed equation represents therefore a conic having an ordinary (two-pointic) contact with the polar conic, and therefore with the curve. It may be added that the two conics intersect besides in a pair of points, and that the line joining these, or common chord of the two conics, is the line represented by the equation $\Pi = 0$; and this being so, the constants (a, b, c) of the line $\Pi = 0$ can be so determined as to give rise to a five-pointic contact.

3. Consider the coordinates of a point of the curve as functions of a single variable parameter; then for the present purpose the coordinates of a point consecutive to (x, y, z) may be taken to be

$$\begin{aligned}x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x, \\y + dy + \frac{1}{2}d^2y + \frac{1}{6}d^3y + \frac{1}{24}d^4y, \\z + dz + \frac{1}{2}d^2z + \frac{1}{6}d^3z + \frac{1}{24}d^4z,\end{aligned}$$

values which, substituted for X, Y, Z , must satisfy the equations

$$\mathbf{T} = 0, \quad D^2U - \Pi \cdot DU = 0.$$

4. I write for shortness

$$\begin{aligned}\partial_1 &= dx \partial_x + dy \partial_y + dz \partial_z, \\ \partial_2 &= d^2x \partial_x + d^2y \partial_y + d^2z \partial_z, \\ \partial_3 &= d^3x \partial_x + d^3y \partial_y + d^3z \partial_z, \\ \partial_4 &= d^4x \partial_x + d^4y \partial_y + d^4z \partial_z,\end{aligned}$$

then the consecutive value of $\mathbf{T}$ is

$$\exp(\partial_1 + \frac{1}{2}\partial_2 + \frac{1}{6}\partial_3 + \frac{1}{24}\partial_4) U$$

(Read $\exp. z$, exponential of z , $= e^z$), which is

$$\begin{aligned}& \times (1 + \partial_1 + \frac{1}{2}\partial_1^2 + \frac{1}{6}\partial_1^3 + \frac{1}{24}\partial_1^4) \\& \times (1 + \frac{1}{2}\partial_2 + \frac{1}{4}\partial_1\partial_2 + \frac{1}{8}\partial_2^2) \\& \times (1 + \frac{1}{6}\partial_3 + \frac{1}{6}\partial_1\partial_3) \\& \times (1 + \frac{1}{24}\partial_4), \\& = U \\& + \partial_1 U \\& + \frac{1}{2}(\partial_1^2 + \partial_2)U \\& + \frac{1}{6}(\partial_1^3 + 3\partial_1\partial_2 + \partial_3)U \\& + \frac{1}{24}(\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2 + \partial_4)U,\end{aligned}$$

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the several terms of which must respectively vanish, and we have therefore

$$\begin{aligned}
 U &= 0, \\
 \partial_1 U &= 0, \\
 \partial_2 U &= -\partial_1^2 U, \\
 \partial_3 U &= -(\partial_1^3 + 3\partial_1 \partial_2)U, \\
 \partial_4 U &= -(\partial_1^4 + 6\partial_1^2 \partial_2 + 4\partial_1 \partial_3 + 3\partial_2^2)U.
 \end{aligned}$$

5. Next, preparing to substitute in the equation

$$D^2 U - \Pi \cdot DU = 0,$$

the consecutive value of DU is

$$(x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x) \partial_x U + \&c.$$

where

$$\partial_x U = (x\partial_x + y\partial_y + z\partial_z)U = mU.$$

Reducing by the above results, the consecutive value of DU is

$$= -\frac{1}{2}\partial_1^2 U - \frac{1}{6}(\partial_1^3 + 3\partial_1 \partial_2)U - \frac{1}{24}(\partial_1^4 + 6\partial_1^2 \partial_2 + 4\partial_1 \partial_3 + 3\partial_2^2)U.$$

6. Hence also writing

$$\begin{aligned}
 P &= ax + by + cz, \\
 \partial_1 P &= a dx + b dy + c dz, \\
 \partial_2 P &= a d^2x + b d^2y + c d^2z,
 \end{aligned}$$

the consecutive value of $-\Pi DU$ is $-(P + \partial_1 P + \frac{1}{2}\partial_2 P)$ multiplied into the consecutive value of DU , and the product is

$$\begin{aligned}
 &= P \cdot \frac{1}{2}\partial_1^2 U \\
 &\quad + P \cdot \frac{1}{6}(\partial_1^3 + 3\partial_1 \partial_2)U \quad + \partial_1 P \cdot \frac{1}{2}\partial_1^2 U \\
 &\quad + P \cdot \frac{1}{24}(\partial_1^4 + 6\partial_1^2 \partial_2 + 4\partial_1 \partial_3 + 3\partial_2^2)U + \partial_1 P \cdot \frac{1}{6}(\partial_1^3 + 3\partial_1 \partial_2)U + \frac{1}{2}\partial_2 P \cdot \frac{1}{2}\partial_1^2 U.
 \end{aligned}$$

7. The consecutive value of $D^2 U$ is

$$\begin{aligned}
 &= (x + dx + \frac{1}{2}d^2x + \frac{1}{6}d^3x + \frac{1}{24}d^4x)^2 \partial_x^2 U + \&c. \\
 &\quad \left. \begin{aligned}
 &= x^2 \\
 &+ 2x dx \\
 &+ x^2 dx + (dx)^2 \\
 &+ \frac{1}{3}x d^3x + dx d^2x \\
 &+ \frac{1}{12}x d^4x + \frac{1}{3}dx d^3x + \frac{1}{4}(d^2x)^2
 \end{aligned} \right\} \partial_x^2 U + \&c.,
 \end{aligned}$$

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which is

$$\left. \begin{aligned} & \partial_1^2 \\ & + 2\partial_1\partial_2 \\ & + \partial_2^2 + \frac{2}{3}\partial_1\partial_3 \\ & + \frac{1}{6}\partial_1\partial_4 + \frac{2}{3}\partial_2\partial_3 + \frac{1}{4}\partial_2^2 \end{aligned} \right\} U;$$

and observing that

$$\begin{aligned} \partial_x^2 U &= m(m-1)U, \\ \partial_x \partial_1 U &= (m-1)\partial_1 U, \\ \partial_x \partial_2 U &= (m-1)\partial_2 U, \\ \partial_x \partial_3 U &= (m-1)\partial_3 U, \\ \partial_x \partial_4 U &= (m-1)\partial_4 U, \end{aligned}$$

and reducing as before, the consecutive value of D^2U is

$$\begin{aligned} &= -(m-2)\partial_1^2 U \\ & - \frac{1}{3}[(m-1)\partial_1^3 + 3(m-2)\partial_1\partial_2]U \\ & - \frac{1}{12}[(m-1)(\partial_1^4 + 6\partial_1^2\partial_2) + (m-2)(4\partial_1\partial_3 + 3\partial_2^2)]U. \end{aligned}$$

8. Substituting in the equation $D^2U - \Pi \cdot DU = 0$, we obtain as the conditions of a five-pointic contact

$$\begin{aligned} & -(m-2)\partial_1^2 U + P \cdot \frac{1}{2}\partial_1^2 U = 0, \\ & -\frac{1}{3}[(m-1)\partial_1^3 + 3(m-2)\partial_1\partial_2]U + P \cdot \frac{1}{6}(\partial_1^3 + 3\partial_1\partial_2)U + \partial_1 P \cdot \frac{1}{2}\partial_1^2 U = 0, \\ & -\frac{1}{12}[(m-1)(\partial_1^4 + 6\partial_1^2\partial_2) + (m-2)(4\partial_1\partial_3 + 3\partial_2^2)]U \\ & + P \cdot \frac{1}{24}[\partial_1^4 + 6\partial_1^2\partial_2 + 4\partial_1\partial_3 + 3\partial_2^2]U + \partial_1 P \cdot \frac{1}{6}(\partial_1^3 + 3\partial_1\partial_2)U + \frac{1}{2}\partial_2 P \cdot \frac{1}{2}\partial_1^2 U = 0; \end{aligned}$$

or reducing

$$\begin{aligned} P &= 2(m-2), \\ \partial_1 P &= \frac{2}{3} \frac{\partial_1^3 U}{\partial_1^2 U}, \\ \partial_2 P &= \frac{1}{2} \frac{[\partial_1^4 + 6\partial_1^2\partial_2]U}{\partial_1^2 U} - \frac{4}{3} \frac{\partial_1^3 U}{\partial_1^2 U} \cdot \frac{[\partial_1^3 + 3\partial_1\partial_2]U}{\partial_1^2 U}, \end{aligned}$$

which are the conditions of a five-pointic contact: it is to be remarked that if only the first and second conditions are satisfied, we have a four-pointic contact, and if only the first condition is satisfied, a three-pointic contact.

9. We have to reduce the last-mentioned equations; suppose that A, B, C are the first derived functions of U , then the equation $\partial_1 U = 0$ may be written

$$A dx + B dy + C dz = 0,$$

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and this will be satisfied identically if

$$\begin{aligned} dx &= B\nu - C\mu, \\ dy &= C\lambda - A\nu, \\ dz &= A\mu - B\lambda, \end{aligned}$$

where λ, μ, ν are arbitrary multipliers, which may be taken to be constants. We have therefore $\partial_1 = \mathbf{D}$, where

$$\mathbf{D} = (B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z.$$

10. The resulting expressions for $\partial_1^2 U$, $\partial_1^3 U$, $\partial_1^4 U$ may be exhibited in the reduced forms given by Hesse, viz. if $\varpi = \lambda x + \mu y + \nu z$, we have

$$\begin{aligned} \partial_1^2 U &= P_2 U - Q_2 \varpi^2, \\ \partial_1^3 U &= P_3 U - Q_3 \varpi^3, \\ \partial_1^4 U &= P_4 U - Q_4 \varpi^4, \end{aligned}$$

where the values of $P_2, P_3, P_4; Q_2, Q_3, Q_4$ are as follows, viz. if (a, b, c, f, g, h) are the second derived functions of U , and if

$$H = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

be the Hessian; if, moreover,

$$\Phi = - \begin{vmatrix} \cdot & \lambda & \mu & \nu \\ \lambda & a & h & g \\ \mu & h & b & f \\ \nu & g & f & c \end{vmatrix}$$

be the bordered Hessian (we may also write $\Phi = (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \lambda, \mu, \nu)^2$, where $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ are the inverse coefficients of (a, b, c, f, g, h) , viz. $\mathfrak{A} = (bc - f^2)$, &c.); and finally, if for shortness we write

$$\Omega = \partial_x \Phi \cdot \partial_x \Phi + \partial_y \Phi \cdot \partial_y \Phi + \partial_z \Phi \cdot \partial_z \Phi,$$

then we have

$$\begin{aligned} P_2 &= \frac{m}{m-1} \Phi, & Q_2 &= \frac{1}{(m-1)^2} H, \\ P_3 &= \frac{m}{m-1} \mathbf{D}\Phi, & Q_3 &= \frac{1}{(m-1)^2} \mathbf{D}H, \\ P_4 &= \frac{m}{m-1} \mathbf{D}^2\Phi - \frac{m\Omega}{(m-1)^2} H, & Q_4 &= \frac{1}{(m-1)^2} \mathbf{D}^2H - \frac{\Omega}{(m-1)^2} \Phi + \frac{3(m-2)}{(m-1)^3} H\Phi. \end{aligned}$$

In the present case $U = 0$, and we have

$$\begin{aligned}\partial_1^2 U &= -Q_2 \varpi^2, \\ \partial_1^3 U &= -Q_3 \varpi^3, \\ \partial_1^4 U &= -Q_4 \varpi^4.\end{aligned}$$

11. Hence, substituting for Q_2 and Q_3 their values, the first and second of the equations for the five-pointic contact give

$$\begin{aligned}P &= 2(m-2), \\ \partial_1 P &= \frac{2}{3} \frac{\mathbf{D}H}{H} \varpi,\end{aligned}$$

and observing that Π is a linear function of (X, Y, Z) , and consequently that $P, \partial_1 P$ denote simply the values which Π assumes when $(x, y, z), (dx, dy, dz)$ are respectively substituted for (X, Y, Z) , we see at once that these two conditions will be satisfied if we put

$$\Pi = \frac{1}{3} \frac{1}{H} DH + \Lambda \cdot DU,$$

where Λ is an arbitrary constant, or, what is the same thing, an arbitrary function of (x, y, z) . We have thus the general equation of a conic of four-pointic contact.

12. The above value of Π gives

$$\partial_2 P = \frac{1}{3} \frac{1}{H} \partial_2 H + \Lambda \partial_1 U,$$

and the third equation of the system of conditions for a five-pointic contact is therefore

$$\frac{1}{3} \frac{1}{H} \partial_2 H + \Lambda \partial_1 U = \frac{1}{2} \frac{[\partial_1^4 + 6\partial_1^2 \partial_2]U}{\partial_1^2 U} - \frac{4}{3} \frac{\partial_1^3 U}{\partial_1^2 U} \cdot \frac{[\partial_1^3 + 3\partial_1 \partial_2]U}{\partial_1^2 U},$$

which leads to the value of Λ .

13. We have in general

$$\begin{aligned}\partial_1^2 U &= \frac{m}{m-1} \Phi U - \frac{1}{(m-1)^2} H \varpi^2, \\ \partial_1^3 U &= \frac{m}{m-1} \mathbf{D}\Phi \cdot U - \frac{1}{(m-1)^2} \mathbf{D}H \cdot \varpi^3, \\ (\partial_1^3 + 3\partial_1 \partial_2) U &= \frac{m}{m-1} \mathbf{D}\Phi \cdot U - \frac{1}{(m-1)^2} \mathbf{D}H \cdot \varpi^3, \\ (\partial_1^4 + 6\partial_1^2 \partial_2) U &= \frac{m}{m-1} \left(2\partial_2 \Phi + \mathbf{D}^2 \Phi + \frac{3}{m-1} \Omega \right) U - \frac{1}{(m-1)^2} \left(\mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi \right) \varpi^4,\end{aligned}$$

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the last two of which have not yet been demonstrated. The value of $\partial_2 \partial_1^2 U$ (which, however, is not required for the present purpose) is

$$= \varpi^2 \frac{-3(m-3)}{m-1} \Omega - \frac{4\Omega}{m-1} \varpi^2,$$

which also is not yet demonstrated.

14. Putting $U = 0$, we have

$$\partial_1^2 U = -\frac{1}{(m-1)^2} H \varpi^2,$$

$$\partial_1^3 U = -\frac{1}{(m-1)^2} \mathbf{D}H \cdot \varpi^3,$$

$$(\partial_1^3 + 3\partial_1 \partial_2) U = -\frac{1}{(m-1)^2} \mathbf{D}H \cdot \varpi^3,$$

$$(\partial_1^4 + 6\partial_1^2 \partial_2) U = -\frac{1}{(m-1)^2} \left(\mathbf{D}^2 H + \mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi \right) \varpi^4,$$

and substituting,

$$\frac{1}{3} \frac{1}{H} \partial_2 H + \frac{1}{(m-1)^2} H \varpi^2 \Lambda = \frac{1}{2} \frac{1}{H} \left(2\partial_2 H + \mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi \right) - \frac{4}{3} \left(\frac{\mathbf{D}H}{H} \right)^2,$$

where the term involving $\partial_2 H$ disappears; the equation may be written

$$\frac{9H \varpi^2}{(m-1)^2} \Lambda = 3H \left(\mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi \right) - 4(\mathbf{D}H)^2,$$

which I will represent by

$$\frac{9H \varpi^2}{(m-1)^2} \Lambda = 3R_2 R_4 - 4R_3^2,$$

the values of R_2, R_3, R_4 being

$$\begin{cases} R_2 = H, \\ R_3 = \mathbf{D}H, \\ R_4 = \mathbf{D}^2 H + \frac{3\Omega}{m-1} \Phi - \frac{3(m-2)}{m-1} H \Phi. \end{cases}$$

15. We have $R_2 = H$, and it will be shown that

$$(m-1)^2 R_3^2 = -9(m-2)^2 H \Psi + 3(m-2) H \Omega \Phi - \Psi \varpi^2,$$

$$(m-1)^2 R_4 = -12(m-2)^2 H \Psi + 4(m-2) \Omega \Phi - \Omega \varpi^2,$$

where for shortness

$$\begin{aligned}\Psi &= (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2, \\ \Omega &= (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H,\end{aligned}$$

and hence, writing

$$9H \varpi^2 \Lambda = 3R_2(m-1)^2 R_4 - 4(m-1)^2 R_3^2,$$

we have

$$9H \varpi^2 \Lambda = -3\Omega\Psi + 4\Psi;$$

or replacing H, Ψ by their values,

$$9H \varpi^2 \Lambda = -3H (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H + 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2,$$

and Λ having this value, the equation of the five-pointic conic is

$$D^2U - \left(\frac{1}{3} \frac{1}{H} DH + \Lambda DU \right) DU = 0,$$

where it will be recollected that the current coordinates are (X, Y, Z) , and that D denotes $X\partial_x + Y\partial_y + Z\partial_z$.

16. I remark, in passing, that the problem of finding the circle of curvature at a given point of a plane curve, is in fact that of determining the conic having with the curve at the given point a three-pointic contact, and besides passing through two given points. The equation of a conic having an ordinary contact, is

$$D^2U - \Pi \cdot DU = 0,$$

where

$$\Pi = aX + bY + cZ,$$

and the condition of a three-pointic contact is

$$ax + by + cz = 2(m-2).$$

Let the coordinates of the two given points be

$$(x_1, y_1, z_1), \quad (x_2, y_2, z_2),$$

and let $(D^2U)_1$ &c. be the corresponding values of D^2U , &c., then we have

$$aX + bY + cZ = \frac{D^2U}{DU},$$

$$ax_1 + by_1 + cz_1 = \left(\frac{D^2U}{DU} \right)_1,$$

$$ax_2 + by_2 + cz_2 = \left(\frac{D^2U}{DU} \right)_2;$$

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and if from the four equations we eliminate a, b, c , we find

$$\begin{vmatrix} XDU, & YDU, & ZDU, & D^2U \\ x, & y, & z, & 2(m-2) \\ x_1, & y_1, & z_1, & \left(\frac{D^2U}{DU}\right)_1 \\ x_2, & y_2, & z_2, & \left(\frac{D^2U}{DU}\right)_2 \end{vmatrix} = 0,$$

for the equation of the conic in question; x, y, z being the coordinates of the point of contact, and X, Y, Z current coordinates.

II. DEMONSTRATION OF IDENTITIES ASSUMED IN THE PRECEDING SECTION.

Proof of the expressions for $(\partial_1^3 + 3\partial_1\partial_2)U$ and $(\partial_1^4 + 6\partial_1^2\partial_2)U$:

17. It will be remembered that ∂_1, ∂_2 stand originally for

$$\begin{aligned} dx \partial_x + dy \partial_y + dz \partial_z, \\ d^2x \partial_x + d^2y \partial_y + d^2z \partial_z, \end{aligned}$$

and that A, B, C being the first derived functions of U , dx, dy, dz are changed into

$$B\nu - C\mu, \quad C\lambda - A\nu, \quad A\mu - B\lambda,$$

and that the resulting value of ∂_1 , viz.

$$(B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z,$$

is also represented by $\mathbf{D}$, so that $\partial_1 = \mathbf{D}$. The corresponding values of d^2x, d^2y, d^2z are

$$\begin{aligned} d^2x &= \nu dB - \mu dC, \\ d^2y &= \lambda dC - \nu dA, \\ d^2z &= \mu dA - \lambda dB, \end{aligned}$$

where we have

$$dA = a dx + h dy + g dz, \quad \&c.,$$

in which dx, dy, dz are to be replaced by the values

$$B\nu - C\mu, \quad C\lambda - A\nu, \quad A\mu - B\lambda;$$

and ∂_2 really denotes what

$$d^2x \partial_x + d^2y \partial_y + d^2z \partial_z$$

becomes when the above values are substituted for d^2x, d^2y, d^2z . But in the expressions $\partial_1^2 U, \partial_1\partial_2 U$ &c., the symbols $\partial_x, \partial_y, \partial_z$ contained in ∂_1 and ∂_2 operate only on U , and not on the variable quantities A, B, C , &c. contained in ∂_1 and ∂_2 .

18. If now we treat ∂_1 as an operand, that is, perform the differentiations on the variable quantities A, B, C which enter into ∂_1 , we obtain

$$\partial_1 \cdot \partial_1 = \partial_1^2 + \partial_2,$$

or, what is the same thing, operating on $\partial_1 U$ with ∂_1 , the result is

$$\partial_1^2 U + \partial_2 U,$$

and in like manner

$$\partial_1 \cdot \partial_1^2 U = \partial_1^3 U + 2\partial_1 \partial_2 U.$$

It is, in fact, upon these principles that Hesse's values of $\partial_1^2 U, \partial_1^3 U, \&c.$ were obtained, and we may by means of them obtain the other expressions assumed in the preceding section.

19. In fact, starting from Hesse's equation,

$$\partial_1^2 U = \frac{m}{m-1} \Phi U - \frac{1}{(m-1)^2} H \varpi^2,$$

we have

$$(\partial_1^3 + 2\partial_1 \partial_2) U = \frac{m}{m-1} (U \mathbf{D} \Phi + \Phi \mathbf{D} U) - \frac{1}{(m-1)^2} (\mathbf{D} H \cdot \varpi^2 + H \cdot 2\varpi \mathbf{D} \varpi).$$

But we have identically $\mathbf{D} U = 0, \mathbf{D} \varpi = 0$, and this equation becomes therefore

$$(\partial_1^3 + 2\partial_1 \partial_2) U = \frac{m}{m-1} U \mathbf{D} \Phi - \frac{1}{(m-1)^2} \mathbf{D} H \cdot \varpi^2;$$

this is precisely Hesse's value of $\partial_1^3 U$, and thus we have $\partial_1 \partial_2 U = 0$, and therefore

$$\partial_1^3 U = \frac{m}{m-1} U \mathbf{D} \Phi - \frac{1}{(m-1)^2} \mathbf{D} H \cdot \varpi^3.$$

20. In like manner, starting from the expression of $\partial_1^3 U$, we have

$$(\partial_1^4 + 3\partial_1^2 \partial_2) U = \frac{m}{m-1} (U \mathbf{D} \cdot \mathbf{D} \Phi + \mathbf{D} U \cdot \mathbf{D} \Phi) - \frac{1}{(m-1)^2} (\mathbf{D} \cdot \mathbf{D} H \cdot \varpi^3 + 3\mathbf{D} H \cdot \varpi^2 \mathbf{D} \varpi),$$

or since $\mathbf{D} U$ and $\mathbf{D} \varpi$ vanish identically, and the values of $\mathbf{D} \cdot \mathbf{D} \Phi, \mathbf{D} \cdot \mathbf{D} H$ are $\partial_1^2 \Phi + \mathbf{D}^2 \Phi, \partial_2 H + \mathbf{D}^2 H$, we have

$$(\partial_1^4 + 3\partial_1^2 \partial_2) U = \frac{m}{m-1} U (\partial_2 \Phi + \mathbf{D}^2 \Phi) - \frac{1}{(m-1)^2} (\partial_2 H + \mathbf{D}^2 H) \varpi^3,$$

and if from the double of this equation we subtract Hesse's equation,

$$\partial_1^4 U = \frac{m}{m-1} \left(U \mathbf{D}^2 \Phi + \frac{3\Omega}{m-1} U \Phi \right) - \frac{1}{(m-1)^2} \left(\mathbf{D}^2 H \cdot \varpi^4 + \frac{3\Omega}{m-1} \Phi \varpi^4 - \frac{3(m-2)}{m-1} H \Phi \varpi^4 \right),$$

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we find the required relation,

$$6\partial_1^2\partial_2U = \frac{m}{m-1}\left(2U\partial_2\Phi + \frac{3\Omega}{m-1}U\Phi\right) - \frac{1}{(m-1)^2}\left(2\partial_2H + \frac{3\Omega}{m-1}\Phi - \frac{3(m-2)}{m-1}H\Phi\right)\varpi^2.$$

Proof of the expression for ∂_2H :

21. We have

$$\partial_2H = \partial_xH(\nu dB - \mu dC) + \partial_yH(\lambda dC - \nu dA) + \partial_zH(\mu dA - \lambda dB),$$

where

$$\begin{aligned} dA &= a dx + h dy + g dz, \\ dB &= h dx + b dy + f dz, \\ dC &= g dx + f dy + c dz, \end{aligned}$$

in which dx, dy, dz are to be replaced by their values: we have therefore

$$\partial_2H = \partial_xH \left\{ \nu[(bC-fB)\lambda + (fA-hC)\mu + (hB-bA)\nu] - \mu[(fC-cB)\lambda + (cA-gC)\mu + (gB-fA)\nu] \right\} + \&c.,$$

where the coefficient of ∂_xH is

$$b\lambda^2 + (gC - cA)\mu^2 + (hB - bA)\nu^2 - (2fA - gB - hC)\mu\nu + (bC - fB)\nu\lambda + (cB - fC)\lambda\mu;$$

or, since we have

$$\begin{aligned} (m-1)A &= ax + hy + gz, \\ (m-1)B &= hx + by + fz, \\ (m-1)C &= gx + fy + cz, \end{aligned}$$

the coefficient, omitting the factor $\frac{1}{m-1}$, which will be afterwards restored, is

$$\begin{aligned} &+ [g(gx + fy + cz) - c(ax + hy + gz)]\mu^2 \\ &+ [h(hx + by + fz) - b(ax + hy + gz)]\nu^2 \\ &+ [2f(ax + hy + gz) - g(hx + by + fz) - h(gx + fy + cz)]\mu\nu \\ &+ [b(gx + fy + cz) - f(hx + by + fz)]\nu\lambda \\ &+ [c(hx + fy + cz) - f(gx + fy + cz)]\lambda\mu \\ &= 0\lambda^2 \\ &+ (-\mathfrak{C}x + \mathfrak{F}y)\mu^2 \\ &+ (-\mathfrak{C}x + \mathfrak{G}z)\nu^2 \\ &+ (-2\mathfrak{F}x + \mathfrak{G}y + \mathfrak{H}z)\mu\nu \\ &+ (-\mathfrak{G}x + \mathfrak{B}z)\nu\lambda \\ &+ (-\mathfrak{H}x + \mathfrak{F}y + \mathfrak{B}z)\lambda\mu, \end{aligned}$$

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which is equal to

$$-(\mathfrak{A}\lambda^2 + \mathfrak{B}\mu^2 + \mathfrak{C}\nu^2 + 2\mathfrak{F}\mu\nu + 2\mathfrak{G}\nu\lambda + 2\mathfrak{H}\lambda\mu)x + (\mathfrak{A}\lambda + \mathfrak{H}\mu + \mathfrak{G}\nu)(\lambda x + \mu y + \nu z).$$

The coefficients of $\partial_y H$, $\partial_z H$ have a similar form; and uniting the three terms, observing that $x\partial_x H + y\partial_y H + z\partial_z H$ is equal to $3(m-2)H$, and attending to the values of Φ , Ω , ϖ , we have, restoring the omitted factor,

$$\partial_2 H = -\frac{3(m-2)}{m-1} H\Phi + \frac{\Omega}{m-1} \varpi.$$

Proof of the expressions for $(DH)^2$ and $D^2\Phi$:

22. These are obtained (for the particular case $m = 4$, which makes but little difference) in Mr Salmon's *Higher Plane Curves*, pp. 88 and 89, and I merely reproduce his investigation; we have

$$(DH)^2 = \{(B\nu - C\mu)\partial_x H + (C\lambda - A\nu)\partial_y H + (A\mu - B\lambda)\partial_z H\}^2,$$

or, what is the same thing,

$$(DH)^2 = \{\lambda(C\partial_y - B\partial_z)H + \mu(A\partial_z - C\partial_x)H + \nu(B\partial_x - A\partial_y)H\}^2;$$

and if we consider first the term which contains λ^2 , the coefficient is

$$[(C\partial_y - B\partial_z)H]^2.$$

Now making use of the equations

$$\begin{aligned} (m-1)A &= ax + hy + gz, \\ (m-1)B &= hx + by + fz, \\ (m-1)C &= gx + fy + cz, \end{aligned}$$

and

$$m(m-1)U = ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0,$$

we have

$$\begin{aligned} (m-1)^2 C^2 &= (gx + fy + cz)^2 - c(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= -\mathfrak{B}x^2 + 2\mathfrak{F}xy - \mathfrak{A}y^2, \\ (m-1)^2 BC &= (hx + by + fz)(gx + fy + cz) - f(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= \mathfrak{F}x^2 - \mathfrak{G}xy - \mathfrak{H}xz + \mathfrak{A}yz, \\ (m-1)^2 B^2 &= (hx + by + fz)^2 - b(ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy) \\ &= -\mathfrak{C}x^2 + 2\mathfrak{G}xz - \mathfrak{A}z^2; \end{aligned}$$

and hence

$$\begin{aligned} (m-1)^2 [(C\partial_y - B\partial_z)H]^2 &= (-\mathfrak{B}x^2 + 2\mathfrak{F}xy - \mathfrak{A}y^2)(\partial_y H)^2 \\ &\quad + 2(-\mathfrak{F}x^2 + \mathfrak{G}xy + \mathfrak{H}xz - \mathfrak{A}yz)\partial_y H \cdot \partial_z H \\ &\quad + (-\mathfrak{C}x^2 + 2\mathfrak{G}xz - \mathfrak{A}z^2)(\partial_z H)^2 \\ &= -x^2(\mathfrak{B}, \mathfrak{C}, \mathfrak{F} \mid \partial_y H, \partial_z H)^2 \\ &\quad + 2x(y\partial_y H + z\partial_z H)(\mathfrak{H}\partial_y H + \mathfrak{G}\partial_z H) \\ &\quad - \mathfrak{A}(y\partial_y H + z\partial_z H)^2, \end{aligned}$$

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and

$$y\partial_y H + z\partial_z H = 3(m-2)H - x\partial_x H,$$

so that the term is

$$\begin{aligned} &= -x^2(\mathfrak{B}, \mathfrak{C}, \mathfrak{F} \mid \partial_y H, \partial_z H)^2 \\ &\quad + 2x(3(m-2)H - x\partial_x H)(\mathfrak{H}\partial_y H + \mathfrak{G}\partial_z H) \\ &\quad - \mathfrak{A}(3(m-2)H - x\partial_x H)^2; \end{aligned}$$

or, reducing, we have

$$\begin{aligned} (m-1)^2[(C\partial_y - B\partial_z)H]^2 &= -x^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 \\ &\quad + 6(m-2)H x (\mathfrak{A}\partial_x H + \mathfrak{H}\partial_y H + \mathfrak{G}\partial_z H) \\ &\quad - 9(m-2)^2 \mathfrak{A} H^2. \end{aligned}$$

23. The other terms may be obtained in a similar manner; and it is easy to see that, collecting all the terms, the sum will be

$$\begin{aligned} &-(\lambda x + \mu y + \nu z)^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 \\ &+ 6(m-2)H(\lambda x + \mu y + \nu z)[(\mathfrak{A}\lambda + \mathfrak{H}\mu + \mathfrak{G}\nu)\partial_x H + (\mathfrak{H}\lambda + \mathfrak{B}\mu + \mathfrak{F}\nu)\partial_y H + (\mathfrak{G}\lambda + \mathfrak{F}\mu + \mathfrak{C}\nu)\partial_z H] \\ &- 9(m-2)^2 H^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \lambda, \mu, \nu)^2; \end{aligned}$$

or, attending to the signification of the symbols $\Psi, \Phi, \Omega, \varpi$, we have

$$(m-1)^2(DH)^2 = -9(m-2)^2 H^2 \Phi + 3(m-2)H \varpi \Omega - \varpi^2 \Psi.$$

24. Next

$$D^2 H = \{(B\nu - C\mu)\partial_x + (C\lambda - A\nu)\partial_y + (A\mu - B\lambda)\partial_z\}^2 H,$$

or, what is the same thing,

$$D^2 H = \{\lambda(C\partial_y - B\partial_z) + \mu(A\partial_z - C\partial_x) + \nu(B\partial_x - A\partial_y)\}^2 H;$$

and if we attend first to the term which contains λ^2 , the coefficient is

$$(C\partial_y - B\partial_z)^2 H.$$

Now substituting for C^2, CB, B^2 as before, we have

$$\begin{aligned} (m-1)^2(C\partial_y - B\partial_z)^2 H &= (-\mathfrak{B}x^2 + 2\mathfrak{F}xy - \mathfrak{A}y^2)\partial_y^2 H \\ &\quad + 2(-\mathfrak{F}x^2 + \mathfrak{G}xy + \mathfrak{H}xz - \mathfrak{A}yz)\partial_y \partial_z H \\ &\quad + (-\mathfrak{C}x^2 + 2\mathfrak{G}xz - \mathfrak{A}z^2)\partial_z^2 H \\ &= -x^2(\mathfrak{B}, \mathfrak{C}, \mathfrak{F} \mid \partial_y, \partial_z)^2 H \\ &\quad + 2x(y\partial_y + z\partial_z)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H \\ &\quad - \mathfrak{A}(y\partial_y + z\partial_z)^2 H, \end{aligned}$$

where it is to be observed that the symbols of differentiation affect H only. We have

$$\begin{aligned} (y\partial_y + z\partial_z)^2 H &= (x\partial_x + y\partial_y + z\partial_z)^2 H \\ &\quad - 2(x\partial_x + y\partial_y + z\partial_z) x\partial_x H \\ &\quad + (x\partial_x)^2 H, \end{aligned}$$

where H is a homogeneous function of the degree $3m - 6$; $x\partial_x H$, since the x is not affected by the differentiation, must be treated as of the degree $3m - 7$, and $(x\partial_x)^2 H$, for the like reason, stands for $x^2\partial_x^2 H$; we have therefore

$$\begin{aligned} (y\partial_y + z\partial_z)^2 H &= (3m - 6)(3m - 7)H \\ &\quad - 2(3m - 7)x\partial_x H \\ &\quad + x^2\partial_x^2 H. \end{aligned}$$

In like manner,

$$\begin{aligned} (y\partial_y + z\partial_z)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H &= (x\partial_x + y\partial_y + z\partial_z)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H - x\partial_x(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H \\ &= (3m - 7)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H - x\partial_x(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H; \end{aligned}$$

and hence

$$\begin{aligned} (m - 1)^2(C\partial_y - B\partial_z)^2 H &= -x^2(\mathfrak{B}, \mathfrak{C}, \mathfrak{F} \mid \partial_y, \partial_z)^2 H \\ &\quad + 2x[(3m - 7)(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H - x\partial_x(\mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H] \\ &\quad - \mathfrak{A}[(3m - 6)(3m - 7)H - 2(3m - 7)x\partial_x H + x^2\partial_x^2 H] \\ &= -x^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H \\ &\quad + 2(3m - 7)x(\mathfrak{A}\partial_x + \mathfrak{H}\partial_y + \mathfrak{G}\partial_z)H \\ &\quad - (3m - 6)(3m - 7)\mathfrak{A}H. \end{aligned}$$

25. The other terms are formed in a similar manner; and collecting all the terms, we have

$$\begin{aligned} (m - 1)^2 D^2 H &= -(3m - 6)(3m - 7)(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \lambda, \mu, \nu)^2 H \\ &\quad + 2(3m - 7)(\lambda x + \mu y + \nu z)[(\mathfrak{A}\lambda + \mathfrak{H}\mu + \mathfrak{G}\nu)\partial_x H + (\mathfrak{H}\lambda + \mathfrak{B}\mu + \mathfrak{F}\nu)\partial_y H + (\mathfrak{G}\lambda + \mathfrak{F}\mu + \mathfrak{C}\nu)\partial_z H] \\ &\quad - (\lambda x + \mu y + \nu z)^2(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H, \end{aligned}$$

or, attending to the signification of the symbols Ψ , Φ , Ω , ϖ , this is

$$(m - 1)^2 D^2 H = -(3m - 6)(3m - 7)H\Phi + (3m - 7)\varpi\Omega - \varpi^2\Psi.$$

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Proof of the expressions for $(m-1)^2 R_3^2$, $(m-1)^2 R_4$:

26. We have

$$\begin{aligned}
 (m-1)^2 R_3^2 &= (m-1)^2 (DH)^2 \\
 &= -9(m-2)^2 H^2 \Phi \\
 &\quad + 3(m-2)H \Omega \varpi \\
 &\quad - \Psi \varpi^2, \\
 (m-1)^2 R_4 &= (m-1)^2 D^2 H + (m-1) \Omega \Phi - (3m-6)(m-1) \Phi H \\
 &= -12(m-2)^2 H \Phi \\
 &\quad + 4(m-2) \Omega \varpi \\
 &\quad - \Omega \varpi^2,
 \end{aligned}$$

which are the expressions required.

Proof of corresponding expressions for $(m-1)^2 Q_3^2$, $(m-1)^2 Q_4$:

27. We have

$$\begin{aligned}
 (m-1)^2 Q_3^2 &= (m-1)^2 (\mathbf{D}H)^2 \\
 &= -9(m-2)^2 H^2 \Phi \\
 &\quad + 3(m-2)H \Omega \Phi \\
 &\quad - \Psi \Phi, \\
 (m-1)^2 Q_4 &= (m-1)^2 \mathbf{D}^2 H - (m-1) \Omega \Phi + (3m-6)(m-1) H \Phi \\
 &= -6(m-2)(m-3) H \Phi \\
 &\quad + 2(m-3) \Omega \Phi \\
 &\quad - \Omega \Phi;
 \end{aligned}$$

to which I join

$$(m-1)^2 Q_2 = H.$$

28. We have consequently

$$\begin{aligned}
 (m-1)^4 (3Q_2 Q_4 - Q_3^2) &= -9(m-2)(m-4) H^2 \Phi \\
 &\quad + 3(m-4) H \Omega \Phi \\
 &\quad + (-3\Omega H + \Psi) \Phi;
 \end{aligned}$$

and

$$(m-1)^6 (3(m-2)Q_2 Q_4 - 2(m-3)Q_3^2) = (-3(m-2)H \Omega + 2(m-3)\Psi) \Phi^2,$$

which for $m=4$ becomes

$$729 (3Q_2 Q_4 - Q_3^2) = (-3\Omega H + \Psi) \Phi^2.$$

29. In the case $m=4$, we have Hesse's theorem, that the equation $3Q_2 Q_4 - Q_3^2 = 0$ gives a curve of the 14th order, which passes through the points of contact of the double tangents, viz. substituting for Ω , Ψ their values, the equation of this curve is

$$\begin{aligned}
 &-3H (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H \\
 &+ (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 = 0.
 \end{aligned}$$

I have added these remarks for the sake of pointing out the striking resemblance of the expressions which occur in the double-tangent problem for $m = 4$, and in the present theory of the five-pointic conic for any value whatever of m . It has not hitherto been shown what the expressions $3(m-2)Q_2Q_4 - 2(m-3)Q_3^2$ and $-3(m-2)H\Omega + 2(m-3)\Psi$ respectively denote, except in the particular case $m = 4$.

III. APPLICATION OF THE FORMULÆ TO THE CUBIC.

30. I shall apply the formula for the five-pointic conic to a cubic; to avoid confusion as to a numerical factor, I write U' , H' in the place of U , H , so that we have

$$\Lambda = \frac{1}{9H'} \left\{ -3(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H' \cdot H' \right. \\ \left. + 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H', \partial_y H', \partial_z H')^2 \right\},$$

and then the equation of the five-pointic conic is

$$D^2U' - \left(\frac{1}{3} \frac{1}{H'} DH' + \Lambda DU' \right) DU' = 0.$$

I take as the equation of the cubic,

$$U = x^3 + y^3 + z^3 + 6lxyz = 0;$$

the formulae Table No. 70 of my Third Memoir on Quantics, *Phil. Trans.*, vol. CXLVI. (1856), pp. 627–647, [144], putting H for HU , give

$$H = l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz.$$

Hence writing

$$U' = \frac{1}{6}(x^3 + y^3 + z^3 + 6lxyz),$$

the first derived functions are

$$\frac{1}{2}(x^2 + 2lyz), \quad \frac{1}{2}(y^2 + 2lzx), \quad \frac{1}{2}(z^2 + 2lxy);$$

the second derived functions, or (a, b, c, f, g, h) , are (x, y, z, lx, ly, lz) , whence $H' = -U$; the inverse coefficients $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ are

$$(yz - l^2x^2, \quad zx - l^2y^2, \quad xy - l^2z^2, \quad l^2yz - lx^2, \quad l^2zx - ly^2, \quad l^2xy - lz^2);$$

and putting $U' = \frac{1}{6}U$ and $H' = -H$, we have

$$\Lambda = -\frac{1}{9H} \left\{ -3(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H \cdot H \right. \\ \left. + 4(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 \right\},$$

and the equation of the five-pointic conic is

$$D^2U - U \left(\frac{1}{3} DH + \frac{1}{6} \Lambda DU \right) DU = 0.$$

31. We have

$$\begin{aligned}
 (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H &= (yz - l^2 x^2) \cdot 6lx \\
 &\quad + (zx - l^2 y^2) \cdot 6ly \\
 &\quad + (xy - l^2 z^2) \cdot 6lz \\
 &\quad + 2(l^2 yz - lx^2) \cdot -(1 + 2l^3)x \\
 &\quad + 2(l^2 zx - ly^2) \cdot -(1 + 2l^3)y \\
 &\quad + 2(l^2 xy - lz^2) \cdot -(1 + 2l^3)z,
 \end{aligned}$$

which is

$$\begin{aligned}
 &= 18l^2 xyz - 6l^4(x^3 + y^3 + z^3) \\
 &\quad - 6l^2(1 + 2l^3)xyz + 2l(1 + 2l^3)(x^3 + y^3 + z^3), \\
 &= (12l^2 - 12l^5)xyz + (2l - 2l^4)(x^3 + y^3 + z^3), \\
 &= 2l(1 - l^3)(x^3 + y^3 + z^3 + 6lxyz),
 \end{aligned}$$

or we have

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H = -2S \cdot U,$$

where S is the quartinvariant (see the Table No. 70). For the present purpose $U = 0$, and consequently

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x, \partial_y, \partial_z)^2 H = 0.$$

32. Next,

$$\begin{aligned}
 (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 &= (yz - l^2 x^2)[3l^2 x^2 - (1 + 2l^3)yz]^2 \\
 &\quad + (zx - l^2 y^2)[3l^2 y^2 - (1 + 2l^3)zx]^2 \\
 &\quad + (xy - l^2 z^2)[3l^2 z^2 - (1 + 2l^3)xy]^2 \\
 &\quad + 2(l^2 yz - lx^2)[3l^2 y^2 - (1 + 2l^3)zx][3l^2 z^2 - (1 + 2l^3)xy] \\
 &\quad + 2(l^2 zx - ly^2)[3l^2 z^2 - (1 + 2l^3)xy][3l^2 x^2 - (1 + 2l^3)yz] \\
 &\quad + 2(l^2 xy - lz^2)[3l^2 x^2 - (1 + 2l^3)yz][3l^2 y^2 - (1 + 2l^3)zx],
 \end{aligned}$$

the first three lines of which are

$$\begin{aligned}
 &9l^4 xyz(x^3 + y^3 + z^3) - 18l^2(1 + 2l^3)x^2 y^2 z^2 + (1 + 2l^3)^2(y^3 z^3 + z^3 x^3 + x^3 y^3) \\
 &\quad - 9l^6(x^5 + y^5 + z^5) + 6l^3(1 + 2l^3)xyz(x^3 + y^3 + z^3) - 3l^2(1 + 2l^3)^2 x^2 y^2 z^2;
 \end{aligned}$$

or collecting and reducing,

$$\begin{aligned}
 &(-9l^6) (x^6 + y^6 + z^6) \\
 &\quad + (1 + 4l^3 + 4l^6)(y^3 z^3 + z^3 x^3 + x^3 y^3) \\
 &\quad + (-15l^4 + 12l^7)(x^3 + y^3 + z^3)xyz \\
 &\quad + (-21l^2 - 48l^5 - 12l^8)x^2 y^2 z^2;
 \end{aligned}$$

the second three lines are

$$18l^4(y^3z^3 + z^3x^3 + x^3y^3) - 12l^2(1 + 2l^3)xyz(x^3 + y^3 + z^3) + 6l^2(1 + 2l^3)^2x^2y^2z^2 \\ - 6l^6x^2y^2z^2 + 12l^3(1 + 2l^3)(y^3z^3 + z^3x^3 + x^3y^3) - 2l(1 + 2l^3)xyz(x^3 + y^3 + z^3);$$

or collecting and reducing,

$$(12l^4 + 42l^6)(y^3z^3 + z^3x^3 + x^3y^3) \\ + (-2l - 20l^4 - 32l^7)(x^3 + y^3 + z^3)xyz \\ + (6l^2 - 30l^5 + 24l^8)x^2y^2z^2.$$

Hence in the first part replacing the top line by

$$-9l^6(x^6 + y^6 + z^6) + 18l^6(y^3z^3 + z^3x^3 + x^3y^3),$$

and uniting the two parts, we find

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 = (1 + 8l^3)^2 (y^3z^3 + z^3x^3 + x^3y^3) \\ + (-9l^6) (x^6 + y^6 + z^6) \\ + (-2l - 5l^4 - 20l^7) (x^3 + y^3 + z^3) xyz \\ + (-15l^2 - 78l^5 + 12l^8) x^2y^2z^2;$$

and referring to the Table No. 70, and writing Θ for ΘU , we have

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \partial_x H, \partial_y H, \partial_z H)^2 = \Theta,$$

where Θ is the first of the three functions which may be chosen to represent the octicovariant of the cubic.

33. We have thus

$$\Lambda = -\frac{4}{9} \frac{\Theta}{H},$$

and thence

$$D^2U - \left(\frac{1}{3} \frac{1}{H} DH - \frac{4}{9} \frac{\Theta}{H} DU \right) DU = 0$$

as the equation of the five-pointic conic: the investigation has been conducted by means of the canonical form of the equation of the cubic, but the form of the result shows that it applies to the equation of the cubic in any form whatever.

34. If, however, we continue to represent the cubic by the canonical equation

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

the result may be further reduced. We have, putting $U = 0$, or writing $x^3 + y^3 + z^3 = -6lxyz$,

$$H = -(1 + 8l^3)xyz;$$

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moreover, putting $U = 0$, the Table No. 70 gives

$$\Theta = (1 + 8l^3)^2(y^3z^3 + z^3x^3 + x^3y^3) - 3l^2x^2y^2z^2;$$

or substituting for H the last-mentioned value, and putting for shortness

$$Q = y^3z^3 + z^3x^3 + x^3y^3 - 3l^2x^2y^2z^2,$$

we have

$$\Theta = (1 + 8l^3)^2 Q,$$

and with these values of H and Q , the equation of the five-pointic conic is

$$D^2U + \left(\frac{1}{3} \frac{DH}{(1 + 8l^3)xyz} + \frac{4}{9} \frac{(1 + 8l^3)Q}{x^2y^2z^2} DU \right) DU = 0,$$

$$DU = 3\{(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z\},$$

$$D^2U = 6\{(X^2 + 2lYZ)x + (Y^2 + 2lZX)y + (Z^2 + 2lXY)z\},$$

or, as it will be convenient to write it,

$$D^2U = 6(x, y, z, lx, ly, lz \mid X, Y, Z)^2,$$

$$DH = \{3l^2x^2 - (1 + 2l^3)yz\}X + \{3l^2y^2 - (1 + 2l^3)zx\}Y + \{3l^2z^2 - (1 + 2l^3)xy\}Z;$$

whence, finally, the equation of the five-pointic conic of the cubic

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0$$

at the point (x, y, z) is

$$\begin{aligned} & 6(1 + 8l^3)x^2y^2z^2(x, y, z, lx, ly, lz \mid X, Y, Z)^2 \\ & + \{(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z\} \\ & \times \left\{ x^2y^2z^2[(3l^2x^2 - (1 + 2l^3)yz)X + (3l^2y^2 - (1 + 2l^3)zx)Y + (3l^2z^2 - (1 + 2l^3)xy)Z] \right. \\ & \quad \left. - Q[(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z] \right\} = 0, \end{aligned}$$

a result which I had previously obtained by a special method.

35. But the expression may be exhibited in a different form by a transformation suggested by a geometrical theorem of Mr Salmon's. In fact the tangent at the point (x, y, z) meets the cubic in the tangential of this point, and the coordinates of the tangential are $x(y^3 - z^3)$, $y(z^3 - x^3)$, $z(x^3 - y^3)$. Calling these ξ , η , ζ , the equation of the tangent to the cubic at the tangential is

$$(\xi^2 + 2l\eta\zeta)X + (\eta^2 + 2l\zeta\xi)Y + (\zeta^2 + 2l\xi\eta)Z = 0.$$

Now we have identically,

$$\begin{aligned} & x^2y^2z^2[(3l^2x^2 - (1 + 2l^3)yz)X + (3l^2y^2 - (1 + 2l^3)zx)Y + (3l^2z^2 - (1 + 2l^3)xy)Z] \\ & - Q[(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z] \\ & = [(\xi^2 + 2l\eta\zeta)X + (\eta^2 + 2l\zeta\xi)Y + (\zeta^2 + 2l\xi\eta)Z] \\ & + U[x^2(-y^3 - z^3 + 2lyz)X + y^2(-z^3 - x^3 + 2lzx)Y + z^2(-x^3 - y^3 + 2lxy)Z]. \end{aligned}$$

In fact this relation will be true if only

$$3x^2y^2z^2(3l^2x^2 - (1 + 2l^3)yz) - Q(x^2 + 2lyz) - (\xi^2 + 2l\eta\zeta) = Ux^2(-y^3 - z^3 + 2lyz);$$

and substituting for ξ , η , ζ and Q their values, the left-hand side is

$$\begin{aligned} & 3x^2y^2z^2(3l^2x^2 - (1 + 2l^3)yz) \\ & - (x^2 + 2lyz)(y^3z^3 + z^3x^3 + x^3y^3 - 3l^2x^2y^2z^2) \\ & - x^2(y^3 - z^3)^2 \\ & - 2lyz(z^3 - x^3)(x^3 - y^3); \end{aligned}$$

and expanding and reducing, the result is

$$x^2(-(y^3 + z^3)x^3 - x^3(y^3 + z^3)) + 2lyz(-2x^3(y^3 + z^3) + x^6) + 12l^2x^2y^2z^2 \cdot x^2;$$

whence, dividing by x^2 , the equation becomes

$$-(y^3 + z^3)(x^3 + y^3 + z^3) + 2lxyz(x^3 - 2(y^3 + z^3)) + 12l^2x^2y^2z^2 = (-y^3 - z^3 + 2lyz)(x^3 + y^3 + z^3 + 6lxyz),$$

which is identically true.

36. Hence in the identical equation putting $U = 0$, we see that the equation of the five-pointic conic may be written

$$\begin{aligned} & 9(1 + 8l^3)x^2y^2z^2(x, y, z, lx, ly, lz \mid X, Y, Z)^2 \\ & + [(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z] \\ & \times [(\xi^2 + 2l\eta\zeta)X + (\eta^2 + 2l\zeta\xi)Y + (\zeta^2 + 2l\xi\eta)Z] = 0, \end{aligned}$$

where ξ , η , ζ stand for $x(y^3 - z^3)$, $y(z^3 - x^3)$, $z(x^3 - y^3)$, the coordinates of the tangential of the given point, and which puts in evidence the geometrical theorem above referred to, viz.

THEOREM.—*The common chord of the five-pointic conic and the polar conic is the tangent to the cubic at the tangential of the given point.*

37. The five-pointic conic meets the cubic in the point of contact, considered as five coincident points, and in a remaining sixth point or point of simple intersection. The process by which I originally obtained the equation of the five-pointic conic, led also to the equation of the line joining the point of contact with the point of simple intersection: the equation of this line is

$$\begin{aligned} & 9x^2y^2z^2 \left\{ \begin{aligned} & Xx((1 + 8l^3)x^4 + (4l + 4l^4)x^2yz + (-2l^2 + 2l^5)y^2z^2) \\ & + Yy((1 + 8l^3)y^4 + (4l + 4l^4)y^2zx + (-2l^2 + 2l^5)z^2x^2) \\ & + Zz((1 + 8l^3)z^4 + (4l + 4l^4)z^2xy + (-2l^2 + 2l^5)x^2y^2) \end{aligned} \right\} \\ & - 9Qx^2y^2z^2 \{ X(3l^2x^2 - (1 + 2l^3)yz) + Y(3l^2y^2 - (1 + 2l^3)zx) + Z(3l^2z^2 - (1 + 2l^3)xy) \} \\ & + Q^2 \{ X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy) \} = 0. \end{aligned}$$

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38. If the conic meet the cubic in six coincident points, that is, if the point of contact be a singular point of the kind already spoken of, or, as we may term it, a sextactic point, then the last-mentioned line must coincide with the tangent at the point. Represent for a moment the equation of the line by

$$AX + BY + CZ = 0,$$

then this line is to coincide with the line

$$(x^2 + 2lyz)X + (y^2 + 2lzx)Y + (z^2 + 2lxy)Z = 0,$$

or we must have

$$B(z^2 + 2lxy) - C(y^2 + 2lzx) = 0,$$

$$C(x^2 + 2lyz) - A(z^2 + 2lxy) = 0,$$

$$A(y^2 + 2lzx) - B(x^2 + 2lyz) = 0,$$

which must be equivalent to a single condition. The terms A, B, C , which contain

$$x^2 + 2lyz, \quad y^2 + 2lzx, \quad z^2 + 2lxy$$

respectively, may, it is clear, be omitted, and omitting also a factor $3x^2y^2z^2$, we may write

$$A = Xx^2yz((1 + 8l^3)x^4 + (4l + 4l^4)x^2yz + (-2l^2 + 2l^5)y^2z^2) - 2Q(3l^2x^2 - (1 + 2l^3)yz),$$

$$B = Xxy^2z((1 + 8l^3)y^4 + (4l + 4l^4)y^2zx + (-2l^2 + 2l^5)z^2x^2) - 2Q(3l^2y^2 - (1 + 2l^3)zx),$$

and the like value for C . The last of the three equations is

$$\begin{aligned} & \{ (y^2 + 2lzx)((1 + 8l^3)x^4 + (4l + 4l^4)x^2yz + (-2l^2 + 2l^5)y^2z^2) \\ & - (x^2 + 2lyz)((1 + 8l^3)y^4 + (4l + 4l^4)y^2zx + (-2l^2 + 2l^5)yz^2x^2) \} \\ & - 2Q\{(y^2 + 2lzx)(3l^2x^2 - (1 + 2l^3)yz) - (x^2 + 2lyz)(3l^2y^2 - (1 + 2l^3)zx)\} = 0, \end{aligned}$$

where the function on the left hand is

$$\begin{aligned} & = 2xyz \left\{ \begin{aligned} & (1 + 8l^3)(x^3y^3 - x^3y^3 + 2l(x^4z - y^4z)) \\ & + (4l + 4l^4)2l(x^3yz^2 - x^3yz^2) \\ & + (-2l^2 + 2l^5)(xy^2z^3 - x^2yz^3) \end{aligned} \right\} \\ & - 2Q\{(1 + 2l^3)(y^3z - x^3z) + 6l^2(x^2z - y^2z)\}, \end{aligned}$$

or, what is the same thing, throwing out the factor z , it is

$$\begin{aligned} & = \{3(1 + 8l^3)x^2y^2 + 6l(1 + 8l^3)xyz(x^2 + y^2) + 30l^2(1 + 8l^3)x^2y^2z^2\}(x^2 - y^2) \\ & - 2Q(1 + 8l^3)(x^3 - y^3); \end{aligned}$$

or throwing out the factor $(1 + 8l^3)$ and substituting for Q its value, it is

$$= \{3x^2y^2 + 6lxyz(x^2 + y^2) + 30l^2x^2y^2z^2 - 2(y^3z^3 + z^3x^3 + x^3y^3 - 3l^2x^2y^2z^2)\}(x^2 - y^2).$$

The first factor, reducing by the equation $x^3 + y^3 + z^3 + 6lxyz = 0$, is

$$\begin{aligned} &= 3x^2y^2 + z^3(x^3 + y^3 + z^3) - 2(y^3z^3 + z^3x^3 + x^3y^3) \\ &= (z^3 - x^3)(z^3 - y^3). \end{aligned}$$

39. Hence putting for the moment

$$M = (y^3 - z^3)(z^3 - x^3)(x^3 - y^3),$$

it appears that the last of the three equations is $Mz = 0$; the first and second are of course $Mx = 0$ and $My = 0$, and the required condition is $M = 0$, that is,

$$(y^3 - z^3)(z^3 - x^3)(x^3 - y^3) = 0,$$

the equation which, combined with the equation of the curve

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

gives the sextactic points. There are consequently twenty-seven such points, and it is at once seen that these are the points of contact of the tangents to the cubic from the points of inflexion, or, what is the same thing, that the twenty-seven sextactic points form nine groups of three each, such that the three points of a group have for their common tangential one of the nine points of inflexion. In fact, let ω be a cube root (real or imaginary) of unity, the three sextactic points of one of the groups will be given by

$$\begin{cases} x - \omega y = 0, \\ x^3 + y^3 + z^3 + 6lxyz = 0. \end{cases}$$

Now consider the tangential of any one of these points, its coordinates are

$$\begin{aligned} x_1 &= x(y^3 - z^3), \\ y_1 &= y(z^3 - x^3), \\ z_1 &= z(x^3 - y^3); \end{aligned}$$

or, reducing by the equation $x - \omega y = 0$, $x_1 = \omega y(x^3 - z^3)$, $y_1 = -y(x^3 - z^3)$, $z_1 = 0$, or, what is the same thing, $x_1 + \omega y_1 = 0$, $z_1 = 0$; that is, the point (x_1, y_1, z_1) is one of the points of inflexion. This is the construction of the sextactic points obtained by Plücker and Steiner.

40. Reverting to the equation of the line joining the point of contact of the five-pointic conic with the point of simple intersection, this meets the cubic in a third point, and Mr Salmon has shown that this third point is in fact the second tangential (tangential of the tangential) of the point of contact, or, what is the same thing:

THEOREM. *The point of simple intersection of the cubic and the five-pointic conic is the third point of intersection with the cubic, of the line joining the point of contact with the second tangential of this point.*